

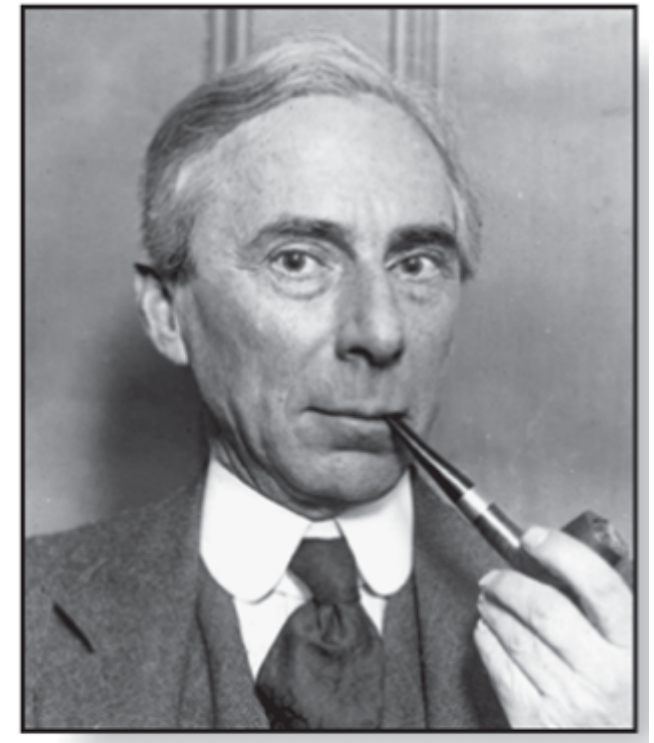
03 Sets and Functions

CS201 Discrete Mathematics

Instructor: Shan Chen

Russell's Paradox

- Let $S = \{x \mid x \notin x\}$ be a set of sets that are **not** members of themselves.
- Paradox:
 - If P is a property, then the set $\{x \mid P(x)\}$ exists (naive set theory): S must exist
 - $S \in S$?
S does not satisfy the property, so $S \notin S$.
 - $S \notin S$?
S is included in the set S, so $S \in S$.
 - $S \in S \leftrightarrow S \notin S$: **S does not exist**
- **Solution:** axiomatic set theory (e.g., Zermelo–Fraenkel)
** out of scope of this course*



Bertrand Russell (1872-1970)
Cambridge, UK
Nobel Prize Winner

Sets

Sets

- A **set** is an **unordered** collection of objects. These objects are called **elements** or **members**.
- Two sets A, B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$.
- Many discrete structures are built with sets:
 - Combinatorics (counting)
 - Relations
 - Graphs
 - ...

Sets

- A **set** is an **unordered** collection of objects. These objects are called **elements** or **members**.
- Examples:
 - $S = \{2, 3, 5, 7\}$
 - $A = \{1, 2, 3, \dots, 100\}$
 - $B = \{a \geq 2 \mid a \text{ is a prime}\}$
 - $C = \{2n \mid n = 0, 1, 2, \dots\}$
- Different ways to represent a set:
 - **Listing** (enumerating) the elements
 - Using **ellipses** “...” if enumeration is hard
 - **Set builder**: $\{x \mid x \text{ has property } P\}$ or $\{x \mid P(x)\}$

Important Sets

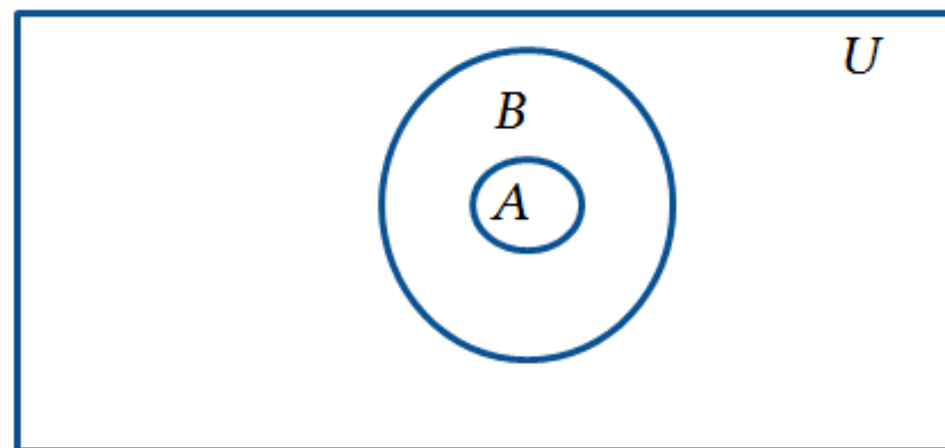
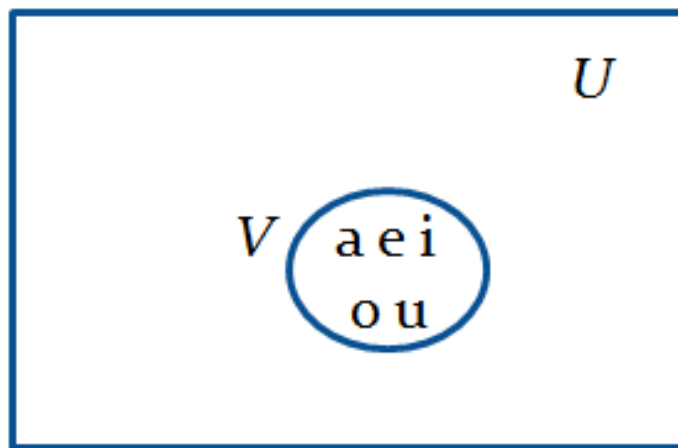
- Natural numbers: $\mathbf{N} = \{ 0, 1, 2, 3, \dots \}$
- Integers: $\mathbf{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$
- Positive integers: $\mathbf{Z}^+ = \{ 1, 2, 3, \dots \}$
- Rational numbers: $\mathbf{Q} = \{ p/q \mid p, q \in \mathbf{Z}, q \neq 0 \}$
- Real numbers: $\mathbf{R}$
- Complex numbers: $\mathbf{C} = \{ a + bi \mid a, b \in \mathbf{R} \}$

Interval Notation

- $[a, b] = \{ x \mid a \leq x \leq b \}$
- $[a, b) = \{ x \mid a \leq x < b \}$
- $(a, b] = \{ x \mid a < x \leq b \}$
- $(a, b) = \{ x \mid a < x < b \}$

Special Sets and Venn Diagrams

- **Universal set 全集:** the set of **all objects** under consideration, denoted by U .
- **Empty set 空集:** the set of **no object**, denoted by $\emptyset$ or $\{\}$.
 - Note that $\emptyset \neq \{\emptyset\}$
- Sets can be visualized using **Venn diagrams**.



John Venn (1834-1923)
Cambridge, UK

Subsets and Proper Subsets

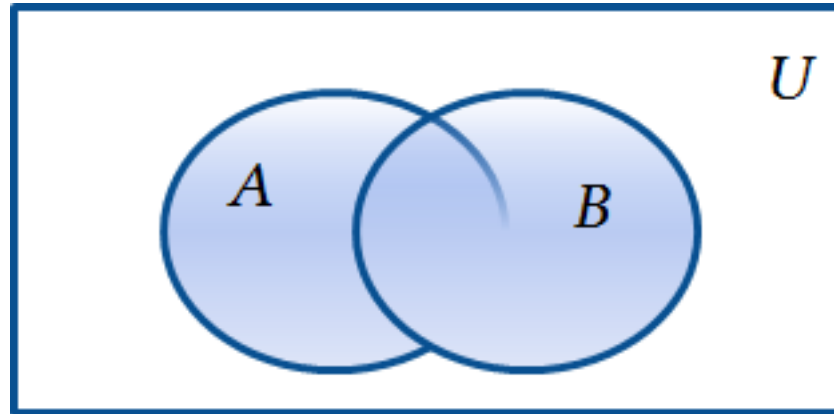
- A set A is called a **subset** 子集 of B , denoted by $A \subseteq B$, if and only if every element of A is an element of B : $\forall x (x \in A \rightarrow x \in B)$
- If $A \subseteq B$ but $A \neq B$, then we say A is a **proper subset** 真子集 of B , denoted by $A \subset B$, i.e., $\forall x (x \in A \rightarrow x \in B) \wedge \exists x (x \in B \wedge x \notin A)$
- Two sets are **equal** if and only if each is a subset of the other
$$A = B \quad \text{if and only if} \quad A \subseteq B \text{ and } B \subseteq A$$
$$\forall x (x \in A \leftrightarrow x \in B) \leftrightarrow (\forall x (x \in A \rightarrow x \in B) \wedge \forall x (x \in B \rightarrow x \in A))$$

Subset Properties

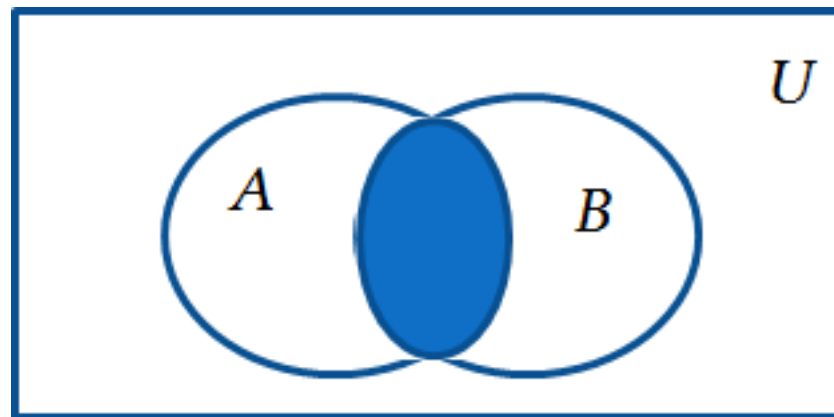
- **Theorem:** $\emptyset \subseteq S$
- Proof: By definition, we need to prove $\forall x (x \in \emptyset \rightarrow x \in S)$.
Since $\emptyset$ does not contain any element, $x \in \emptyset$ is **always false**.
Then the implication is **always true**. * *vacuous proof*
- **Theorem:** $S \subseteq S$
- Proof: By definition, we need to prove $\forall x (x \in S \rightarrow x \in S)$, which is **obviously true**.

Set Operations

- **Union 并**: The union of sets A and B , denoted by $A \cup B$, is the set $\{x \mid x \in A \vee x \in B\}$.

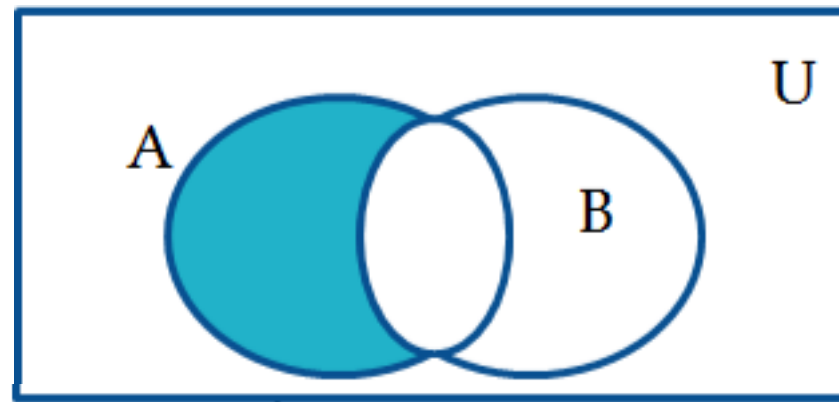


- **Intersection 交**: The intersection of sets A and B , denoted by $A \cap B$, is the set $\{x \mid x \in A \wedge x \in B\}$. Two sets A and B are called **disjoint 互斥** if their intersection is empty, i.e., $A \cap B = \emptyset$.

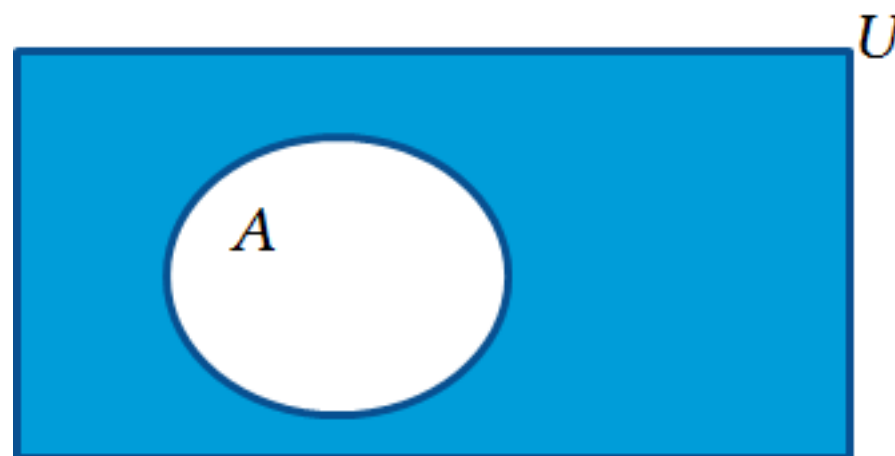


Set Operations

- **Difference 差**: The difference of sets A and B , denoted by $A - B$, is the set that contains all the elements of A that are not in B , i.e., $A - B = \{x \mid x \in A \wedge x \notin B\}$.



- **Complement 补**: The complement of set A (with respect to the universal set U), denoted by $\bar{A}$, is the set $U - A$, i.e., $\bar{A} = \{x \in U \mid x \notin A\}$.



Exercise (1 min)

- Let $U = \{0, 1, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8\}$.
Compute the following:

- $A \cup B$

- $A \cap B$

- $\bar{A}$

- $\bar{B}$

- $A - B$

- $B - A$

Exercise (1 min)

- Let $U = \{0, 1, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8\}$.
Compute the following:

- $A \cup B$ $\{1, 2, \dots, 8\}$
- $A \cap B$ $\{4, 5\}$
- $\bar{A}$ $\{0, 6, 7, 8, 9, 10\}$
- $\bar{B}$ $\{0, 1, 2, 3, 9, 10\}$
- $A - B$ $\{1, 2, 3\}$
- $B - A$ $\{6, 7, 8\}$

Union and Intersection (Generalized)

- **The union of a collection of sets:** the set that contains those elements that are members of **at least one set** in the collection:

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \cdots \cup A_n.$$

- **The intersection of a collection of sets:** the set that contains those elements that are members of **all sets** in the collection:

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \cdots \cap A_n.$$

Set Identities

○ Identity laws 恒等律

- $A \cup \emptyset = A$
- $A \cap U = A$

○ Domination laws 支配律

- $A \cup U = U$
- $A \cap \emptyset = \emptyset$

○ Idempotent laws 幂等律

- $A \cup A = A$
- $A \cap A = A$

○ Commutative laws 交换律

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

○ Associative laws 结合律

- $A \cup (B \cup C) = (A \cup B) \cup C$
- $A \cap (B \cap C) = (A \cap B) \cap C$

○ Distributive laws 分配律

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Set Identities

- Absorption laws 吸收律
 - $A \cup (A \cap B) = A$
 - $A \cap (A \cup B) = A$
- Complement laws 补集律
 - $A \cup \bar{A} = U$
 - $A \cap \bar{A} = \emptyset$
- De Morgan's laws
 - $\overline{A \cap B} = \bar{A} \cup \bar{B}$
 - $\overline{A \cup B} = \bar{A} \cap \bar{B}$
- Complementation laws 双重补集律
 - $\bar{\bar{A}} = A$

How do we prove these laws?

Let's see the first De Morgan's law for example...

Proofs of $\overline{A \cap B} = \bar{A} \cup \bar{B}$

- Using membership tables 隶属表: * *requires tedious calculations*

A	B	$\bar{A}$	$\bar{B}$	$\overline{A \cap B}$	$\bar{A} \cup \bar{B}$
1	1	0	0	0	0
1	0	0	1	1	1
0	1	1	0	1	1
0	0	1	1	1	1

Proofs of $\overline{A \cap B} = \bar{A} \cup \bar{B}$

- Using **set builder** notation and logical equivalences:

$$\overline{A \cap B} = \{x \mid x \in \overline{A \cap B}\}$$

Definition

$$= \{x \mid x \notin A \cap B\}$$

Definition of complement

$$= \{x \mid \neg(x \in (A \cap B))\}$$

Definition

$$= \{x \mid \neg(x \in A \wedge x \in B)\}$$

Definition of intersection

$$= \{x \mid \neg(x \in A) \vee \neg(x \in B)\}$$

De Morgan's

$$= \{x \mid x \notin A \vee x \notin B\}$$

Definition

$$= \{x \mid x \in \bar{A} \vee x \in \bar{B}\}$$

Definition of complement

$$= \{x \mid x \in \bar{A} \cup \bar{B}\}$$

Definition of union

$$= \bar{A} \cup \bar{B}$$

Definition

- Using logical equivalence without set builders: ** less elegant*
 - Prove $\forall x(x \in \overline{A \cap B} \leftrightarrow x \in \bar{A} \cup \bar{B})$ is a tautology. ** see textbook*

Using Set Identities

◦ Show that $\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$

◦ Proof:
$$\begin{aligned}\overline{A \cup (B \cap C)} &= \bar{A} \cap \overline{(B \cap C)} \\ &= \bar{A} \cap (\bar{B} \cup \bar{C}) \\ &= (\bar{B} \cup \bar{C}) \cap \bar{A} \\ &= (\bar{C} \cup \bar{B}) \cap \bar{A}\end{aligned}$$

De Morgan's

De Morgan's

Commutative

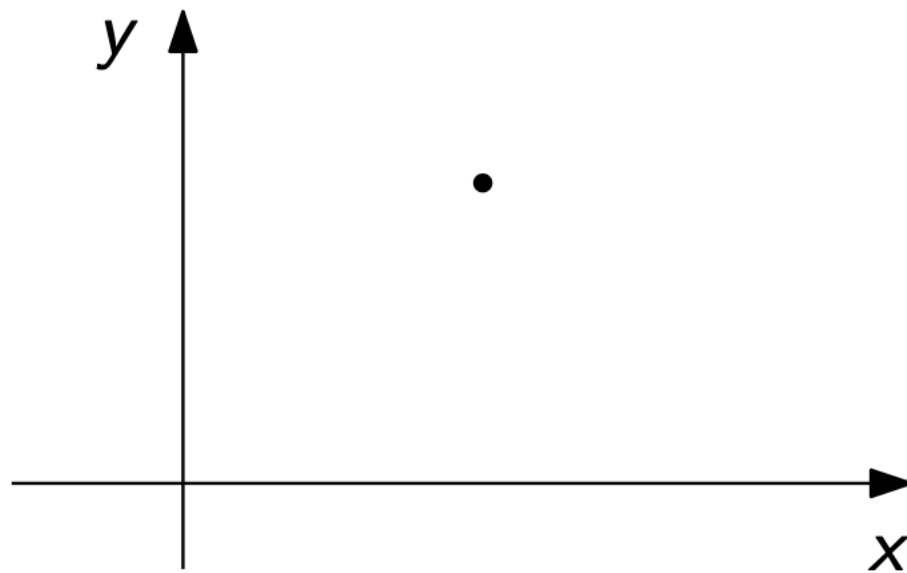
Commutative

Cardinality

- Let S be a set. If there are **exactly n distinct elements** in S , where n is a **nonnegative** integer, we say that S is a **finite set** and n is the **cardinality 基数** of S , denoted by $|S|$.
- A set S is **infinite** if it is **not finite**.
- Examples:
 - $A = \{ 1, 2, 3, \dots, 20 \}$ $|A| = 20$
 - $B = \{ 1, 2, 3, \dots \}$ **infinite**
 - $\emptyset$ $|\emptyset| = 0$
- **Cardinality of the union:** $|A \cup B| = |A| + |B| - |A \cap B|$
 - $|A \cap B|$ **counted twice** in $|A| + |B|$
 - Known as the **inclusion-exclusion principle** for 2 sets

Tuples

- An n -tuple n 元组 $(a_1, a_2, \dots, a_n)$ is an **ordered** collection that has a_1 as its first element, a_2 as its second element, and so on, until a_n as its last element.
- Example: coordinates of a point in the 2-D plane are 2-tuples



Cartesian Product

- Let A, B be sets. The Cartesian product 笛卡尔乘积 of A and B , denoted by $A \times B$, is the set of all 2-tuples (a, b) , for $a \in A, b \in B$:

$$A \times B = \{ (a, b) \mid a \in A \wedge b \in B \}$$

- Example: $A = \{ 1, 2 \}, B = \{ a, b, c \}$
 - $A \times B = \{ (1, a), (1, b), (1, c), (2, a), (2, b), (2, c) \}$
- Properties:
 - $A \times B \neq B \times A$ * *order matters*
 - $|A \times B| = |A| \times |B|$ if A, B are finite sets

Cartesian Product (Generalized)

- In general, the Cartesian product of sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is defined as:

$$A_1 \times A_2 \times \dots \times A_n = \{ (a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n \}$$

- Example: $A = \{ 0, 1 \}$, $B = \{ 1, 2 \}$, $C = \{ 0, 1, 2 \}$
 - $A \times B \times C = \{ (0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2) \}$

Power Sets

- Given a set S , the **power set** 幂集 of S is the **set of all subsets** of the set S , denoted by $\mathcal{P}(S)$.
- Examples:
 - $\emptyset$ $\mathcal{P}(\emptyset) = \{\emptyset\}$
 - $\{1\}$ $\mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}$
 - $\{1, 2\}$ $\mathcal{P}(\{1,2\}) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$
 - $\{1, 2, 3\}$ $\mathcal{P}(\{1,2,3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\}$
- If S is a set with $|S| = n$, then $|\mathcal{P}(S)| = ?$
 - $|\mathcal{P}(S)| = 2^n$ *Hint: each element is either in the subset or not in it*

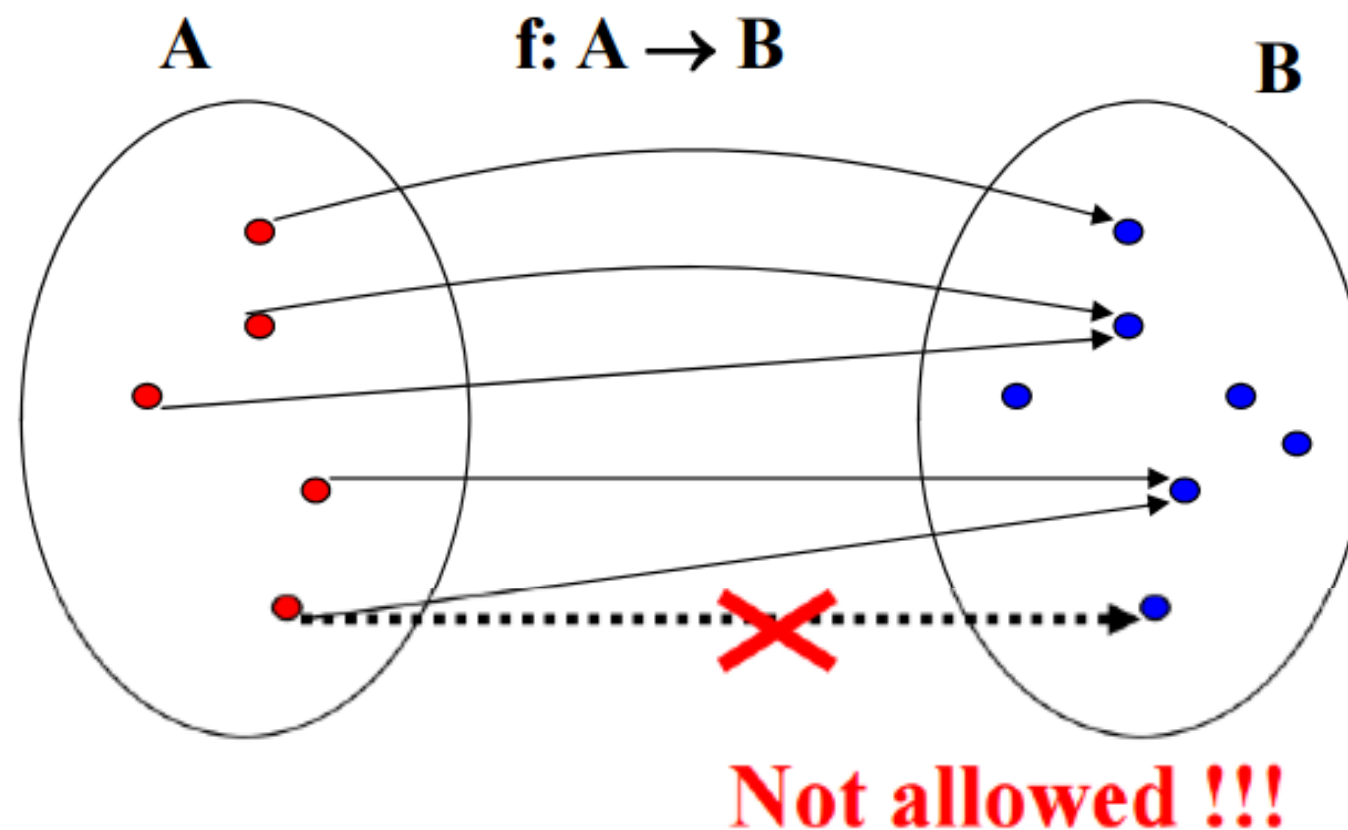
Computer Representation of Sets

- How to store sets (which are subsets of the same universal set) in a computer?
 - **Naive solution:** explicitly store the elements of each set in a **list**
 - **Better solution:** assign a **bit** in a bit string to **each element** in the universal set and **set the bit to 1 if the element is in the set and set it to 0 if otherwise**
 - Example: $U = \{ 1, 2, 3, 4, 5 \}$, $A = \{ 2, 5 \}$, $B = \{ 1, 5 \}$
 - **Sets as bit strings:** $A = 01001$, $B = 10001$
 - **Union:** $A \text{ or } B = 11001 = \{ 1, 2, 5 \}$
 - **Intersection:** $A \text{ and } B = 00001 = \{ 5 \}$
 - **Complement:** $A \text{ xor } 11111 = 10110 = \{ 1, 3, 4 \}$
- * set operations converted to bitwise operations of Boolean algebra*

Functions

Functions

- Let A and B be two sets. A **function** 函数 from A to B , denoted by $f: A \rightarrow B$, is an assignment of **exactly one element of B** to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .
 - Also called a **mapping** 映射 or **transformation** 变换.



Representing Functions

- Representing functions $f : A \rightarrow B$:
 - explicitly state the assignments between elements from A to B
 - use a formula
- Examples:
 - $A = \{ 1, 2, 3 \}, B = \{ a, b, c \}$

f is defined as $1 \mapsto c, 2 \mapsto a, 3 \mapsto c$. Is f a function?
Yes

g is defined as $1 \mapsto c, 1 \mapsto b, 2 \mapsto a, 3 \mapsto c$. Is g a function?
No
 - $A = \{ 0, 1, \dots, 9 \}, B = \{ 0, 1, 2 \}$

h is defined as $h(x) = x \bmod 3$. Is h a function?
Yes

Important Sets of Functions

- Let f be a function from A to B . We say A is the **domain** 定义域 of f and B is the **codomain** 培域 of f . If $f(a) = b$, b is called the **image** 像 of a and a is a **preimage** 原像 of b . The **range** 值域 of f is the set of all images of elements of A , denoted by $f(A)$. We also say that f maps A to B .
- Example: $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$ shown in the following figure
 - the **image** of 1 is c
 - 2 is a **preimage** of a
 - the **domain** of f is $\{1, 2, 3\}$
 - the **codomain** of f is $\{a, b, c\}$
 - the **range** of f is $\{a, c\}$

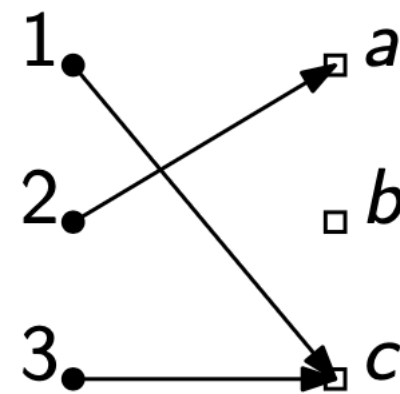
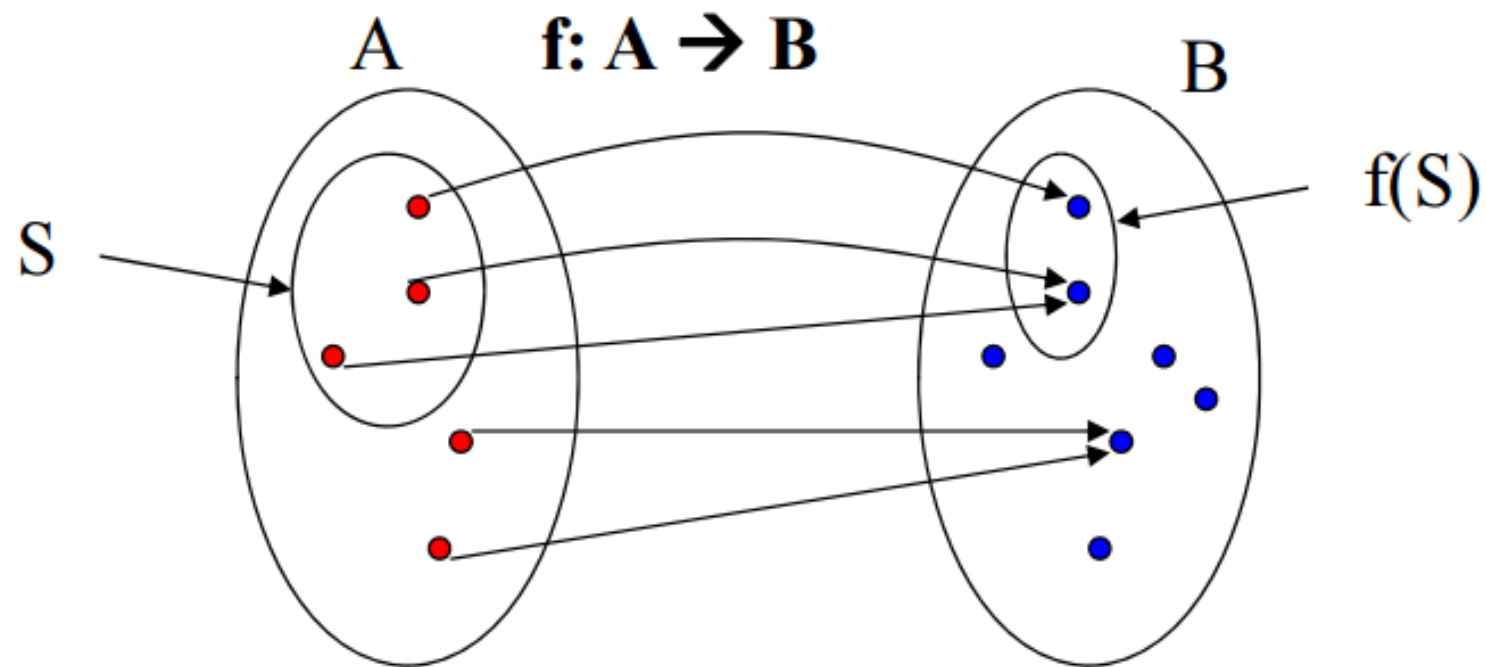
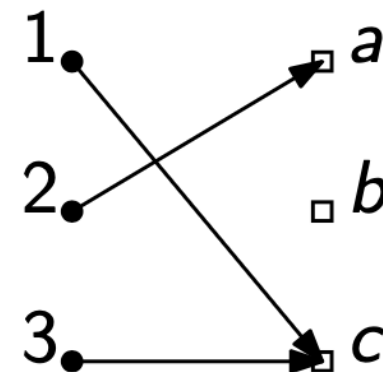


Image of a Subset

- For a function $f : A \rightarrow B$ and $S \subseteq A$, the image of S is a subset of B that consists of the images of elements in S , denoted by $f(S)$, i.e., $f(S) = \{ f(x) \mid x \in S \}$.

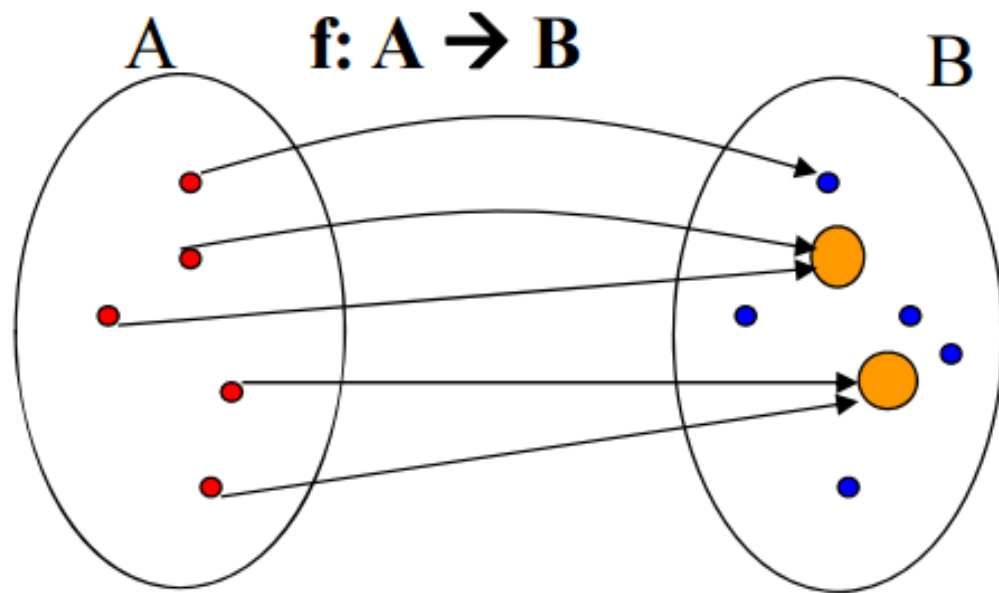


- Example: Let $S = \{ 1, 3 \}$, what is $f(S)$?
 - $f(S) = \{c\}$

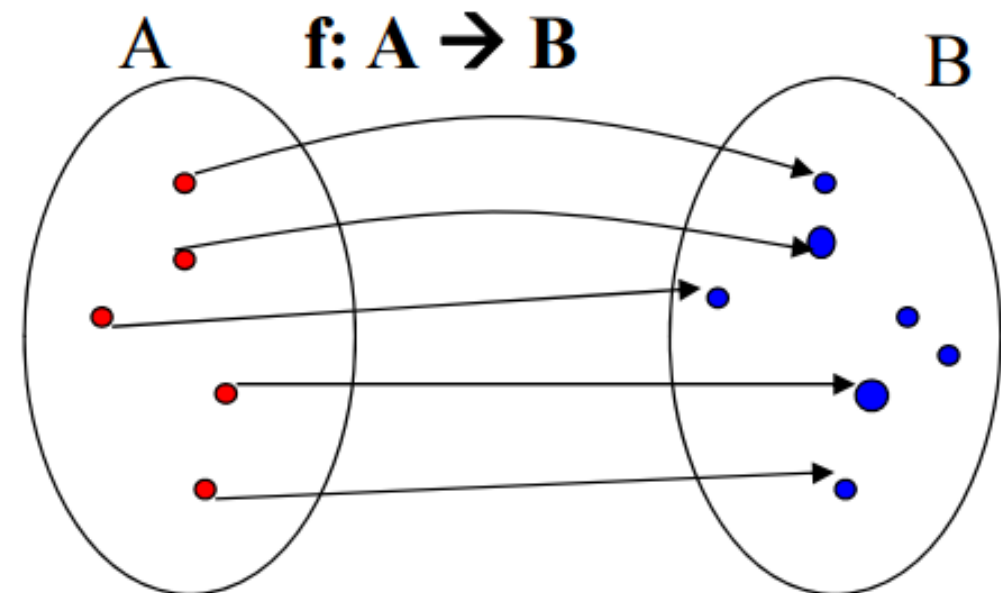


Injective (One-to-One) Functions

- A function f is called **one-to-one** or **injective** 单射, if and only if $f(x) = f(y)$ implies $x = y$ for all x, y in the domain of f . In this case, f is called an **injection**.
- Alternatively: A function is **one-to-one** or **injective** if and only if $x \neq y$ implies $f(x) \neq f(y)$. ** contrapositive!*



Not injective



Injective function

Injective (One-to-One) Functions

- Examples:

- Let $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$, where $1 \mapsto c, 2 \mapsto a, 3 \mapsto c$. Is f injective?

No

- Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$, where $g(x) = 2x - 1$. Is g one-to-one?

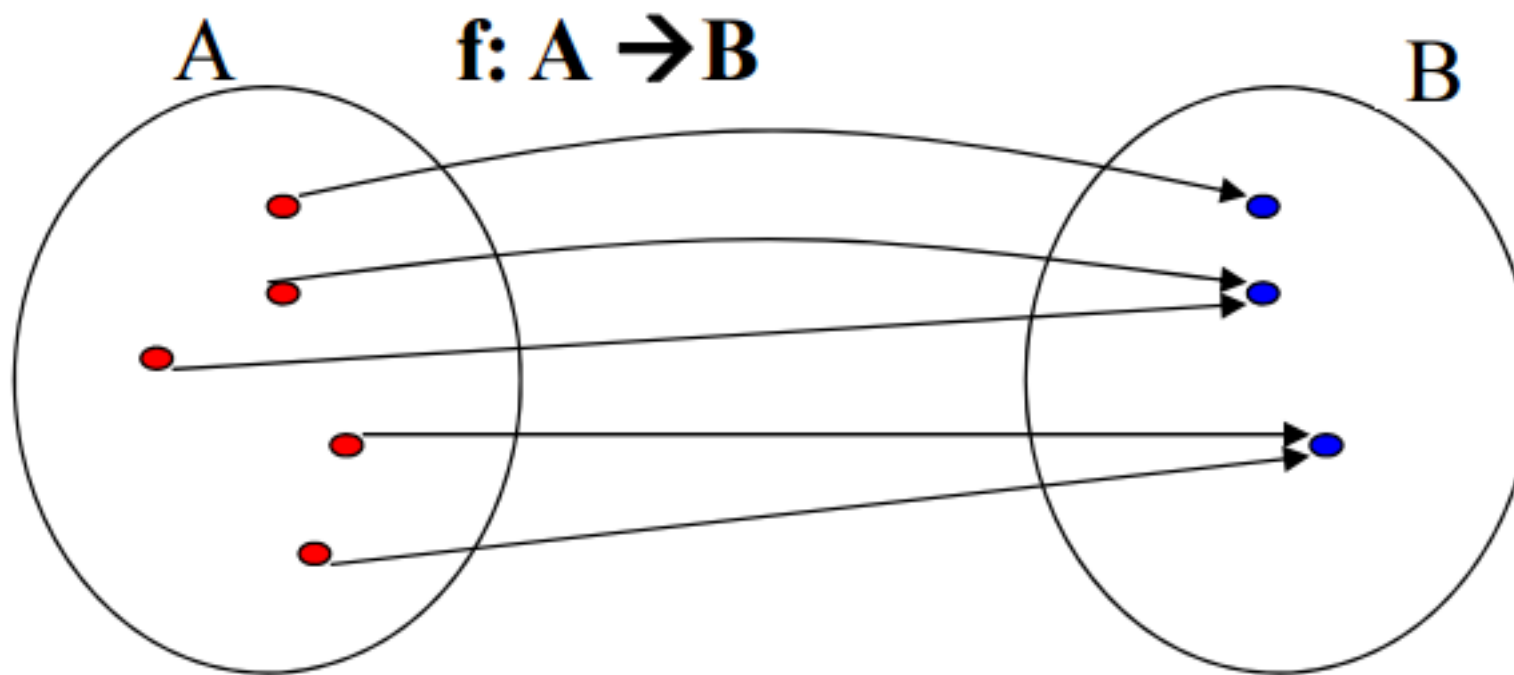
Yes

- Let $h : \mathbb{Z} \rightarrow \mathbb{Z}$, where $h(x) = x^2 + 1$. Is h injective?

No

Surjective (Onto) Functions

- A function f is called **onto** or **surjective** 滿射, if and only if for every $b \in B$ there is an element $a \in A$ such that $f(a) = b$. In this case, f is called a **surjection**.
- Alternatively: A function is **onto** or **surjective** if and only if all codomain elements are covered, i.e., $f(A) = B$.



Surjective (Onto) Functions

- Examples:

- Let $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$, where $1 \mapsto c, 2 \mapsto a, 3 \mapsto c$. Is f onto?

No

- Let $g : \mathbf{Z} \rightarrow \mathbf{Z}$, where $g(x) = 2x - 1$. Is g surjective?

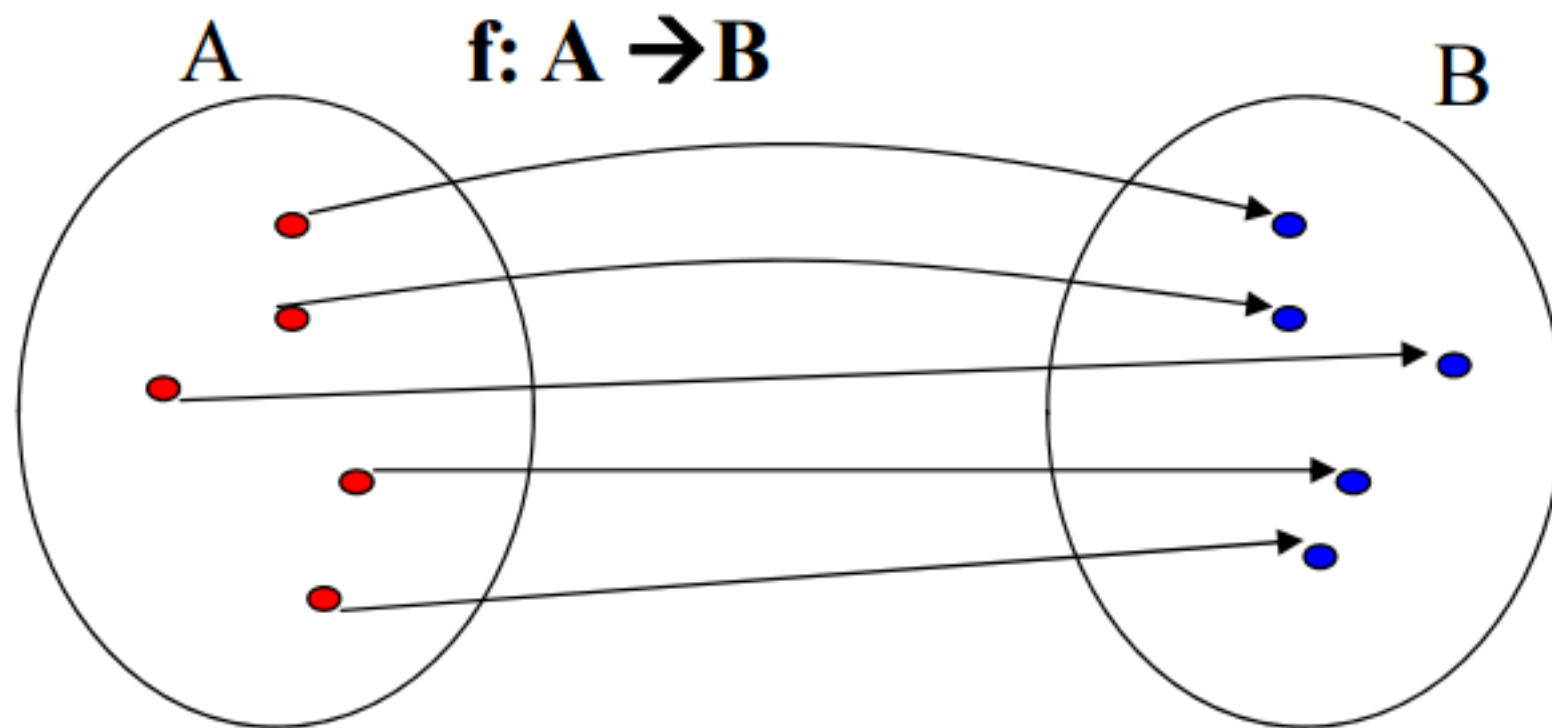
No

- Let $h : \{1, 2, 3, 4\} \rightarrow \{0, 1, 2\}$, where $h(x) = x \bmod 3$. Is h onto?

Yes

Bijjective Functions

- A function f is called **bijjective 双射**, if and only if it is **both one-to-one and onto, i.e., both injective and surjective**. In this case, f is called a **bijection**.
 - Also known as a **one-to-one correspondence 一一对应**.



Bijjective Functions

- Examples:

- Let $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$, where $1 \mapsto c, 2 \mapsto a, 3 \mapsto b$. Is f bijective?

Yes

- Let $g : \mathbf{N} \rightarrow \mathbf{N}$, where $g(x) = \lfloor x/2 \rfloor$ (floor function). Is g bijective?

No (not injective)

- The identity function 恒等函数 on set A is the function $I_A : A \rightarrow A$ so that $I_A(a) = a$ for all $a \in A$, i.e., it assigns each element to itself. Identity functions are bijective functions.

Relevant Proof Strategy

- Consider a function $f: A \rightarrow B$.

To show that f is <i>injective</i> (<i>one-to-one</i>)	Show that for all $x, y \in A$ if $x \neq y$ then $f(x) \neq f(y)$
To show that f is not <i>injective</i>	Find specific $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$
To show that f is <i>surjective</i> (<i>onto</i>)	Show that for all $y \in B$ there exists $x \in A$ such that $f(x) = y$
To show that f is not <i>surjective</i>	Find a specific $y \in B$ such that $f(x) \neq y$ for all $x \in A$

Exercise (2 mins)

- Prove that “for an arbitrary function $f : A \rightarrow B$ with $|A| = |B| = n$, f is one-to-one if and only if f is onto.”

Hint: prove “if” and “only if”

To show that f is <i>injective</i> (<i>one-to-one</i>)	Show that for all $x, y \in A$ if $x \neq y$ then $f(x) \neq f(y)$
To show that f is not <i>injective</i>	Find specific $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$
To show that f is <i>surjective</i> (<i>onto</i>)	Show that for all $y \in B$ there exists $x \in A$ such that $f(x) = y$
To show that f is not <i>surjective</i>	Find a specific $y \in B$ such that $f(x) \neq y$ for all $x \in A$

Exercise (2 mins)

- Prove that “for an arbitrary function $f : A \rightarrow B$ with $|A| = |B| = n$, f is one-to-one if and only if f is onto.”

Hint: prove “if” and “only if”

- Proof:

- “only if” part: Suppose that f is one-to-one. Let’s do a direct proof.

Let $A = \{x_1, x_2, \dots, x_n\}$. Since f is one-to-one, $f(x_i) \neq f(x_j)$ for $i \neq j$. Therefore, $|f(A)| = |\{f(x_1), \dots, f(x_n)\}| = n$. Since $|B| = n$ and $f(A) \subseteq B$, we have $f(A) = B$.

- “if” part: Suppose that f is onto. Let’s use proof by contradiction.

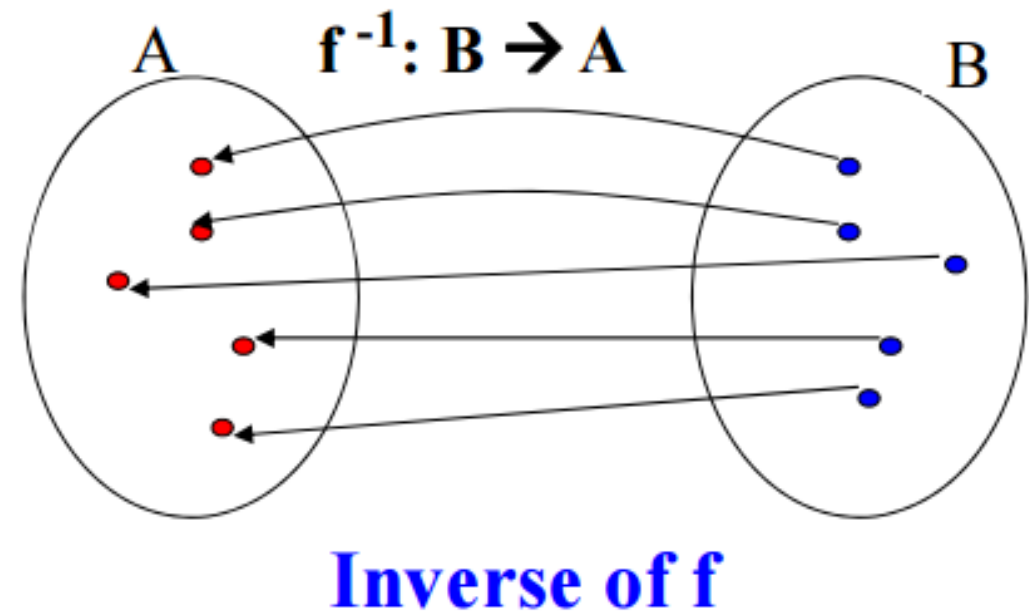
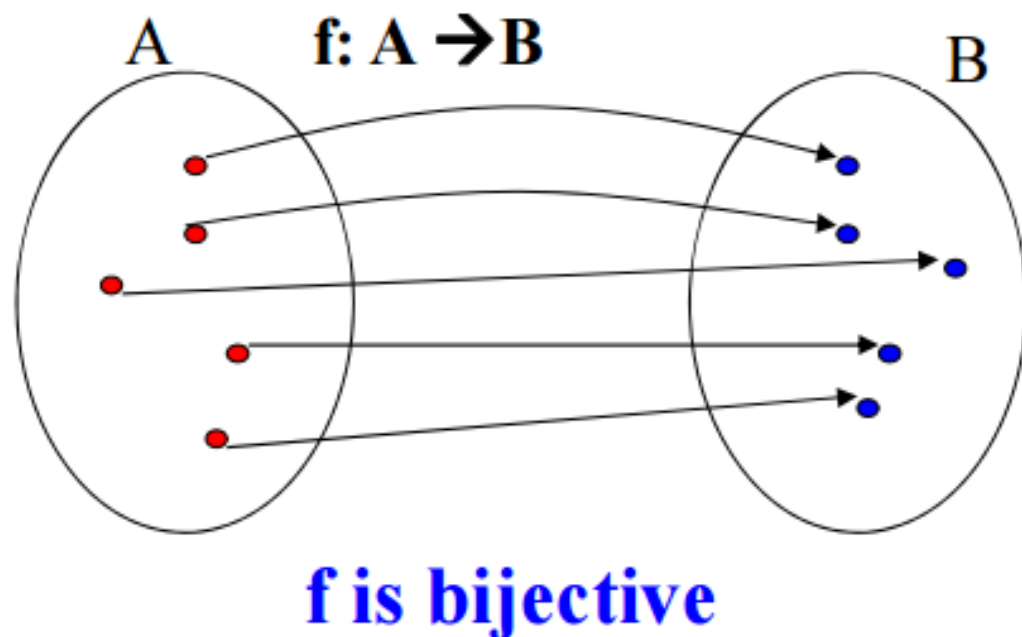
Let $A = \{x_1, x_2, \dots, x_n\}$. If f is not injective, then there exist $x_i \neq x_j$ such that $f(x_i) = f(x_j)$. Then, $|f(A)| = |\{f(x_1), \dots, f(x_n)\}| \leq n - 1$. However, this contradicts with “ f is onto” (i.e., $f(A) = B$, which implies $|f(A)| = |B| = n$). Therefore, f is one-to-one.

Note

- **Claim:** For an arbitrary function $f: A \rightarrow A$, f is one-to-one if and only if f is onto. * *what about this claim? still true?*
- **No!** Set A must be finite for the above **Claim** to be true.
 - Counterexample: $f: \mathbf{N} \rightarrow \mathbf{N}$, $f(x) = 2x$. Here f is one-to-one but not onto, e.g., 1 has no preimage.

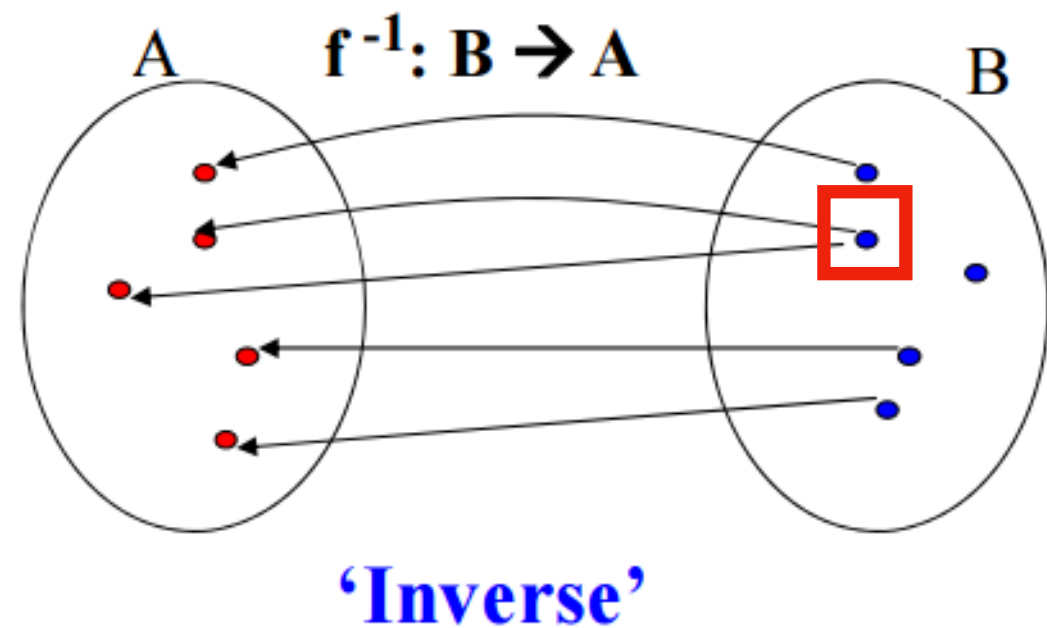
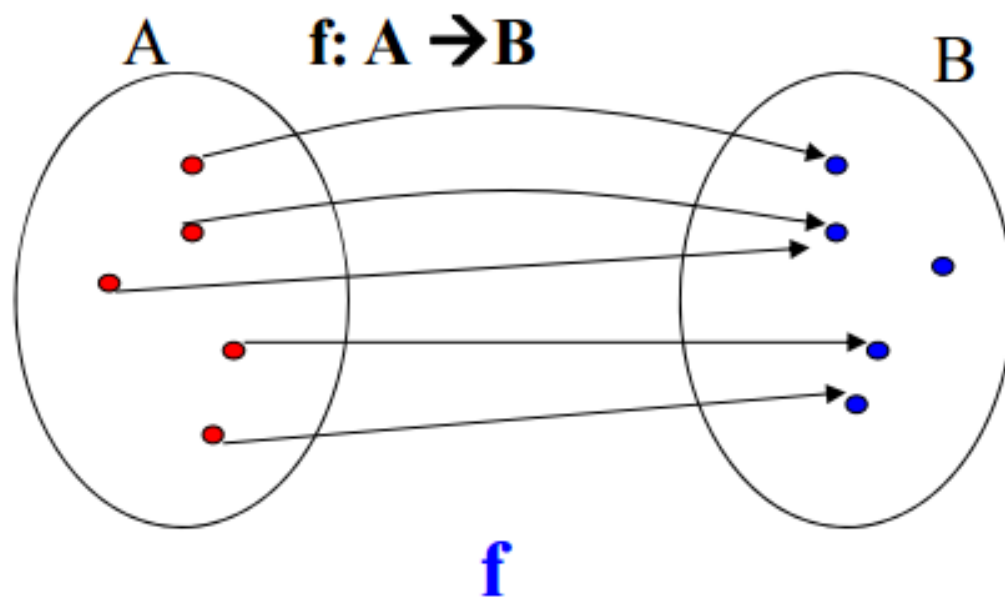
Inverse Functions

- Let $f: A \rightarrow B$ be a **bijection**. The **inverse 逆** of f is the function that assigns to $b \in B$ the **unique** element $a \in A$ such that $f(a) = b$, denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$. In this case, f is called **invertible 可逆**.



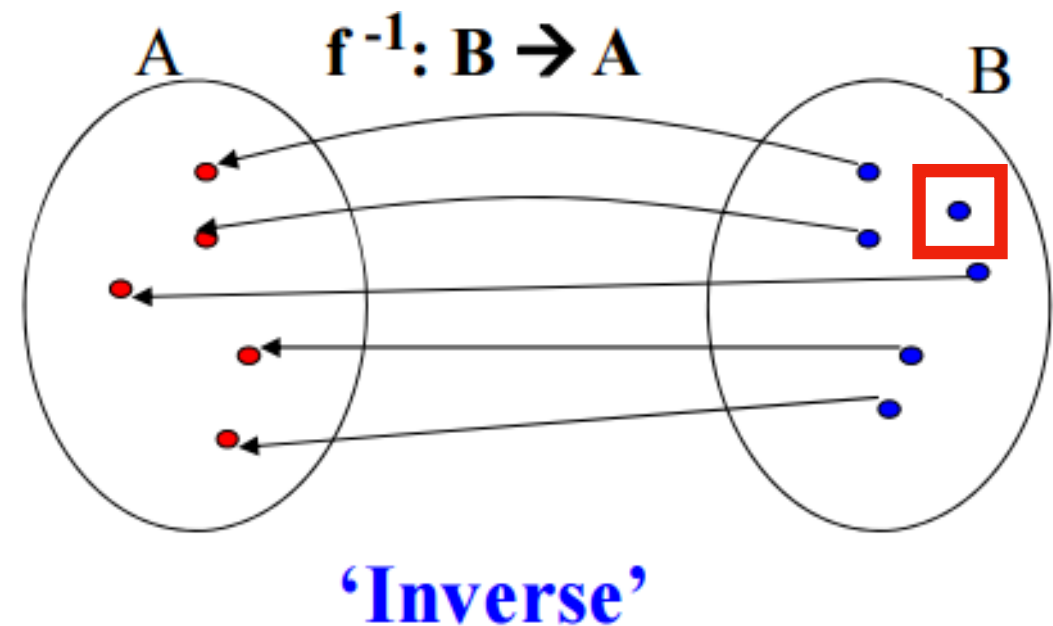
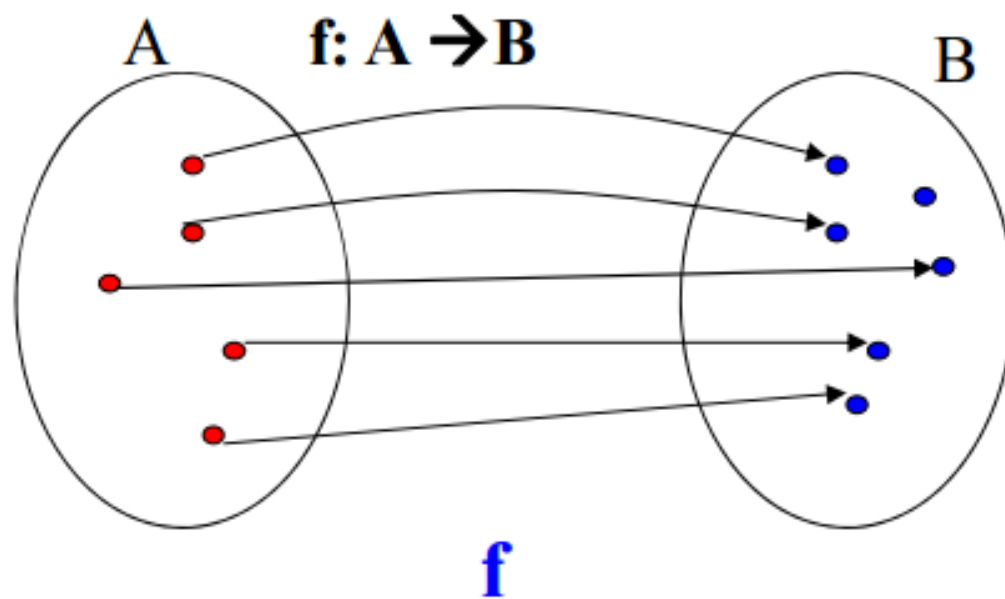
Inverse Functions

- **Theorem:** If f is **not a bijection**, then it is **impossible** to define the inverse function of f .
- Proof by cases:
 - **Case 1:** f is **not injective**
The inverse is **not a function**: at least one element of B is **mapped to two different** elements of A



Inverse Functions

- **Theorem:** If f is **not a bijection**, then it is **impossible** to define the inverse function of f .
- Proof by cases:
 - **Case 2:** f is **not surjective**
The inverse is **not a function**: at least one element of B is **not mapped to any** element of A



Inverse Functions

- Example 1:

- $f: \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = 2x - 1$
- What is the inverse function f^{-1} ?

$$f^{-1}(x) = (x + 1)/2$$

- Example 2:

- $f: \mathbf{Z} \rightarrow \mathbf{Z}$, where $f(x) = 2x - 1$
- Is f invertible?

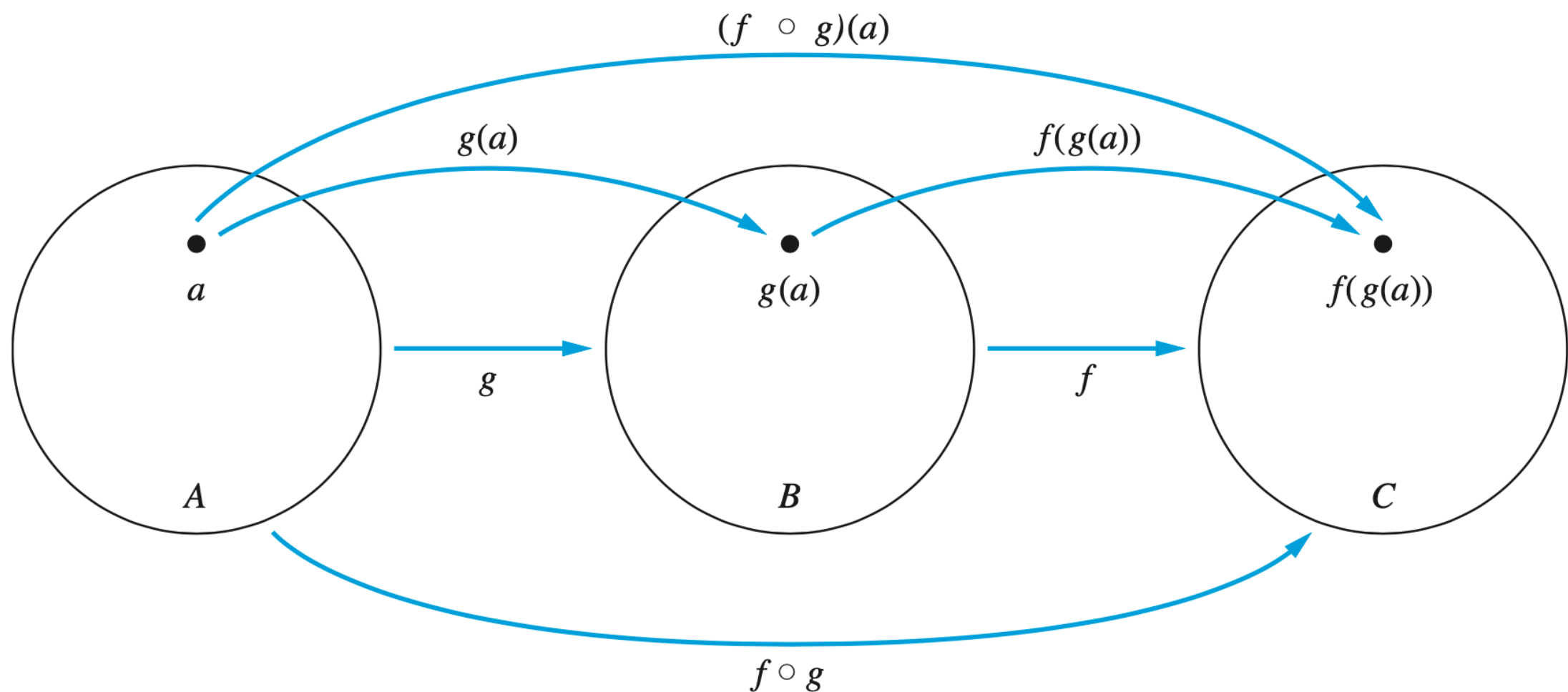
No, because f is **not onto**, e.g., 2 has no preimage.

Operations of Real-Valued Functions

- Let f_1 and f_2 be functions from A to $\mathbf{R}$. Their sum $f_1 + f_2$ and product $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$:
 - $(f_1 + f_2)(x) = f_1(x) + f_2(x)$
 - $(f_1 f_2)(x) = f_1(x) f_2(x)$
- Example: $f_1 = x - 1$, $f_2 = x^3 + 1$
 - $(f_1 + f_2)(x) = (x - 1) + (x^3 + 1) = x^3 + x$
 - $(f_1 f_2)(x) = (x - 1)(x^3 + 1) = x^4 - x^3 + x - 1$

Composition of Functions

- Consider two functions $f: B \rightarrow C$ and $g: A \rightarrow B$.
The **composition** 合成 of the functions f and g , denoted by $f \circ g$, is defined as $(f \circ g)(x) = f(g(x))$.



Composition of Functions

○ Example 1: $A = \{1, 2, 3\}$ and $B = \{a, b, c, d\}$

- $f : A \rightarrow B$ where $1 \mapsto b, 2 \mapsto a, 3 \mapsto d$

- $g : A \rightarrow A$ where $1 \mapsto 3, 2 \mapsto 1, 3 \mapsto 2$

- What is $f \circ g$?

$f \circ g : A \rightarrow B$ where $1 \mapsto d, 2 \mapsto b, 3 \mapsto a$

○ Example 2:

- $f : \mathbf{Z} \rightarrow \mathbf{Z}$ where $f(x) = 2x$

- $g : \mathbf{Z} \rightarrow \mathbf{Z}$ where $g(x) = x^2$

- What are $f \circ g$ and $g \circ f$?

$(f \circ g)(x) = 2x^2$ $(g \circ f)(x) = 4x^2$ * order of composition matters

Composition of Functions

- **Theorem:** Suppose f is a **bijection** from A to B and let I_A and I_B respectively denote the **identity functions** on the sets A and B . Then,
 - $f^{-1} \circ f = I_A$
 - $f \circ f^{-1} = I_B$
- **Proof:** Consider any a, b such that $f(a) = b$
 - $(f^{-1} \circ f)(a) = f^{-1}(f(a)) = f^{-1}(b) = a$
 - $(f \circ f^{-1})(b) = f(f^{-1}(b)) = f(a) = b$

Some Important Functions

- The **floor** 向下取整 function assigns a real number x the **largest** integer that is $\leq x$, denoted by $\lfloor x \rfloor$.
- The **ceiling** 向上取整 function assigns a real number x the **smallest** integer that is $\geq x$, denoted by $\lceil x \rceil$.
- The **factorial** 阶乘 function f assigns a non-negative integer the **product of the first n** positive integers, denoted by $f(n) = n!$.
 - $0! = 1!/1 = 1$

TABLE 1 Useful Properties of the Floor and Ceiling Functions.

(n is an integer, x is a real number)

$$(1a) \quad \lfloor x \rfloor = n \text{ if and only if } n \leq x < n + 1$$

$$(1b) \quad \lceil x \rceil = n \text{ if and only if } n - 1 < x \leq n$$

$$(1c) \quad \lfloor x \rfloor = n \text{ if and only if } x - 1 < n \leq x$$

$$(1d) \quad \lceil x \rceil = n \text{ if and only if } x \leq n < x + 1$$

$$(2) \quad x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$$

$$(3a) \quad \lfloor -x \rfloor = -\lceil x \rceil$$

$$(3b) \quad \lceil -x \rceil = -\lfloor x \rfloor$$

$$(4a) \quad \lfloor x + n \rfloor = \lfloor x \rfloor + n$$

$$(4b) \quad \lceil x + n \rceil = \lceil x \rceil + n$$

Exercise (2 mins)

- Prove that “if x is a real number, then $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 1/2 \rfloor$.”

Hint: notice that $x = \lfloor x \rfloor + y$ for $0 \leq y < 1$ and prove it by cases

TABLE 1 Useful Properties of the Floor and Ceiling Functions.

(n is an integer, x is a real number)

(1a) $\lfloor x \rfloor = n$ if and only if $n \leq x < n + 1$

(1b) $\lceil x \rceil = n$ if and only if $n - 1 < x \leq n$

(1c) $\lfloor x \rfloor = n$ if and only if $x - 1 < n \leq x$

(1d) $\lceil x \rceil = n$ if and only if $x \leq n < x + 1$

(2) $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$

(3a) $\lfloor -x \rfloor = -\lceil x \rceil$

(3b) $\lceil -x \rceil = -\lfloor x \rfloor$

(4a) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$

(4b) $\lceil x + n \rceil = \lceil x \rceil + n$

Exercise (2 mins)

- Prove that “if x is a real number, then $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 1/2 \rfloor$.”

Hint: notice that $x = \lfloor x \rfloor + y$ for $0 \leq y < 1$ and prove it by cases

- Proof by cases:

- By definition of floor function, $x = \lfloor x \rfloor + y$ where $0 \leq y < 1$.

- **Case 1: $0 \leq y < 1/2$**

We have $0 \leq 2y < 1$ and $0 \leq y + 1/2 < 1$, so

$$\lfloor 2x \rfloor = \lfloor 2\lfloor x \rfloor + 2y \rfloor = 2\lfloor x \rfloor + \lfloor 2y \rfloor = 2\lfloor x \rfloor$$

$$\lfloor x + 1/2 \rfloor = \lfloor \lfloor x \rfloor + y + 1/2 \rfloor = \lfloor x \rfloor + \lfloor y + 1/2 \rfloor = \lfloor x \rfloor$$

- **Case 2: $1/2 \leq y < 1$**

We have $1 \leq 2y < 2$ and $1 \leq y + 1/2 < 2$, so

$$\lfloor 2x \rfloor = \lfloor 2\lfloor x \rfloor + 2y \rfloor = 2\lfloor x \rfloor + \lfloor 2y \rfloor = 2\lfloor x \rfloor + 1$$

$$\lfloor x + 1/2 \rfloor = \lfloor \lfloor x \rfloor + y + 1/2 \rfloor = \lfloor x \rfloor + \lfloor y + 1/2 \rfloor = \lfloor x \rfloor + 1$$

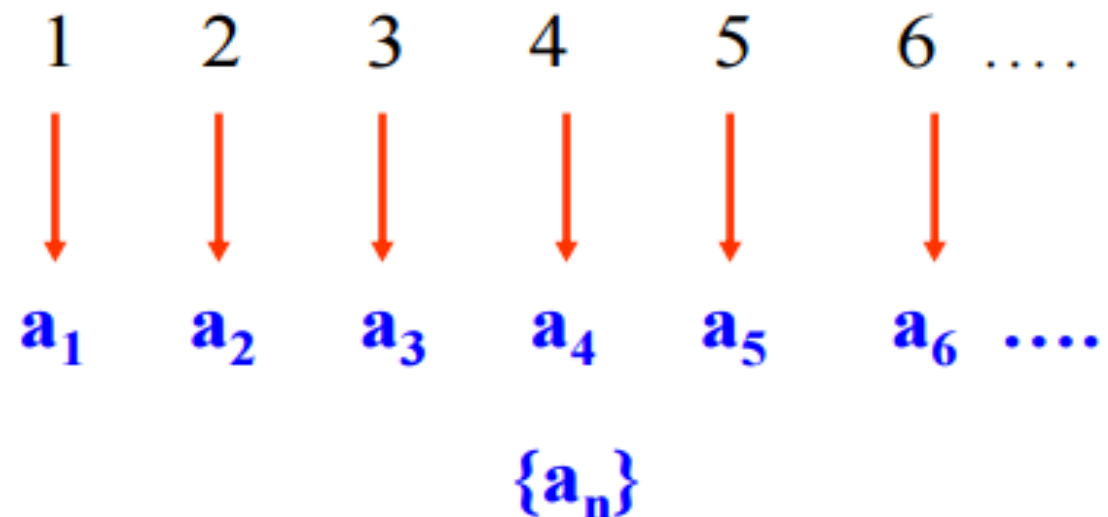
Assignment 2

- Deadline for Assignment 2: Oct 21
- Requirements:
 - Write your solutions in English (otherwise 50% penalty).
 - Submit a single PDF file on Blackboard (otherwise 20% penalty).
- DO NOT CHEAT! Plagiarism will be punished severely.
 - It's OK to form study groups and share ideas, but do not plagiarize!
- Reminder: Please submit your Assignment 1 by Sep 30.

Sequences and Summations

Sequences

- A **sequence** is a **function** from a subset of the set of integers (usually $\{0, 1, 2, \dots\}$ or $\{1, 2, 3, \dots\}$) to a set S .



- Notations:
 - a_n denotes the image of the integer n
 - $\{a_n\}$ denotes the sequence $a_0, a_1, a_2, \dots$ or $a_1, a_2, a_3, \dots$
** note that here $\{a_n\}$ is not a set!*

Defining Sequences

- A sequence $\{a_n\}$ can be defined by providing explicit formulas for their terms, also called closed formulas 闭合式.
- Examples:
 - $a_n = n^2$, where $n = 1, 2, 3, \dots$
 - $a_n = (-1)^n$, where $n = 0, 1, 2, \dots$
- Alternatively, the n -th element a_n can be defined recursively 递归 in terms of one or more previous elements. Such a recurrence relation together with initial conditions defines sequence $\{a_n\}$.
- Examples:
 - $a_n = a_{n-1} + 2$ for $n \geq 1$ and $a_0 = 1$
 - $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$ and $f_0 = 0, f_1 = 1$ * Fibonacci sequence

Arithmetic/Geometric Progression

- **Arithmetic progression 等差数列:** a sequence of the form

$$a, a + d, a + 2d, \dots, a + nd, \dots$$

where the **initial term a** and **common difference d** are real numbers.

- Example: $a_n = -1 + 4n$, where $n = 0, 1, 2, 3, \dots$

- **Geometric progression 等比数列:** a sequence of the form

$$a, ar, ar^2, \dots, ar^n, \dots$$

where the **initial term a** and **common ratio r** are real numbers.

- Example: $a_n = 3 \cdot \left(\frac{1}{2}\right)^n$, where $n = 0, 1, 2, 3, \dots$

Summations

- The summation of terms of a sequence is denoted by

$$\sum_{j=m}^n a_j = a_m + a_{m+1} + \cdots + a_n$$

- The variable j is referred to as the index of summation 求和指标 and the choice of the letter j is arbitrary.
 - m is the lower limit 下限 of the summation
 - n is the upper limit 上限 of the summation
- Useful summation identities:

$$\sum_{j=m}^n (ax_j + by_j) = a \sum_{j=m}^n x_j + b \sum_{j=m}^n y_j \quad \sum_{i=1}^m \sum_{j=1}^n a_i b_j = \sum_{i=1}^m a_i \sum_{j=1}^n b_j = \sum_{j=1}^n b_j \sum_{i=1}^m a_i$$

Summations

- The sum from the *0-th* term to the *n-th* term of the arithmetic progression $a, a + d, a + 2d, \dots, a + nd$ is

$$\sum_{j=0}^n (a + jd) = (n + 1)a + d \sum_{j=0}^n j = (n + 1)a + d \frac{n(n + 1)}{2}$$

- The sum from the *0-th* term to the *n-th* term of the geometric progression $a, ar, ar^2, \dots, ar^n$ is

$$\sum_{j=0}^n (ar^j) = a \sum_{j=0}^n r^j = a \frac{r^{n+1} - 1}{r - 1}$$

*What about the sum from the *m-th* term to the *n-th* term?*

Summations

- The sum from the m -th term to the n -th term of the arithmetic progression $a + md, a + (m + 1)d, \dots, a + nd$ is

$$\sum_{j=m}^n (a + jd) = (n - m + 1)a + d \frac{(m + n)(n - m + 1)}{2}$$

- The sum from the m -th term to the n -th term of the geometric progression $ar^m, ar^{m+1}, \dots, ar^n$ is

$$\sum_{j=m}^n (ar^j) = a \sum_{j=m}^n r^j = a \frac{r^{n+1} - r^m}{r - 1}$$

Hint: can be proved directly or using $\sum_{j=m}^n = \sum_{j=0}^n - \sum_{j=0}^{m-1}$

Examples

○ Calculate the following summations:

$$\diamond S = \sum_{j=1}^5 (2 + 3j) \quad 55$$

$$\diamond S = \sum_{j=3}^5 (2 + 3j) \quad 42$$

$$\diamond S = \sum_{i=1}^4 \sum_{j=1}^2 (2i - j) \quad 28$$

$$\diamond S = \sum_{j=0}^3 2(5)^j \quad 312$$

$$\diamond S = \sum_{i=1}^4 \sum_{j=1}^3 ij \quad 60$$

$$\sum_{j=m}^n (a + jd) = (n - m + 1)a + d \frac{(m + n)(n - m + 1)}{2} \quad \sum_{j=m}^n (ar^j) = a \frac{r^{n+1} - r^m}{r - 1}$$

Infinite Series

- An infinite geometric series 无穷几何级数 can be computed in the closed form for $|x| < 1$.

$$\sum_{k=0}^{\infty} x^k = \lim_{n \rightarrow \infty} \sum_{k=0}^n x^k = \lim_{n \rightarrow \infty} \frac{x^{n+1} - 1}{x - 1} = \frac{0 - 1}{x - 1} = \frac{1}{1 - x}$$

- Differentiating the above formula on both sides:

$$\sum_{k=1}^{\infty} kx^{k-1} = \sum_{k=0}^{\infty} (k+1)x^k = \frac{1}{(1-x)^2}$$

** proved true for $|x| < 1$ by a calculus theorem about infinite series*

- Proof without calculus:

$$\text{Let } S_n = 1 + 2x + \dots + nx^{n-1}$$

$$(1-x)S_n = S_n - xS_n = 1 + x + \dots + x^{n-1} - nx^n = (1-x^n)/(1-x) - nx^n$$

$$S_n = (1-x^n)/(1-x)^2 - nx^n/(1-x) \rightarrow 1/(1-x)^2 \text{ (if } n \rightarrow \infty) \text{ * L'Hôpital's}$$

Useful Summation Formulas

TABLE 2 Some Useful Summation Formulae.	
<i>Sum</i>	<i>Closed Form</i>
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

Cardinality of Infinite Sets

Cardinality of Sets

- Recall that the **cardinality** 基数 of a finite set S is defined by the number of the elements in S , denoted by $|S|$.
- **Definition:** Sets A and B have the **same cardinality** if there is a **one-to-one correspondence (bijection)** between A and B , denoted by $|A| = |B|$.
 - Cardinality of **infinite** sets may be **counter-intuitive**, e.g., $|\mathbf{N}| = |\mathbf{Z}|$.
- **Definition:** If there exists a **injective** function from A to B , then we say the cardinality of A is **less than or equal to** the cardinality of B , denoted by $|A| \leq |B|$. Moreover, if $|A| \leq |B|$ and A and B have **different** cardinalities, we say that the cardinality of A is **less than** the cardinality of B , denoted by $|A| < |B|$.

Schröder-Bernstein Theorem

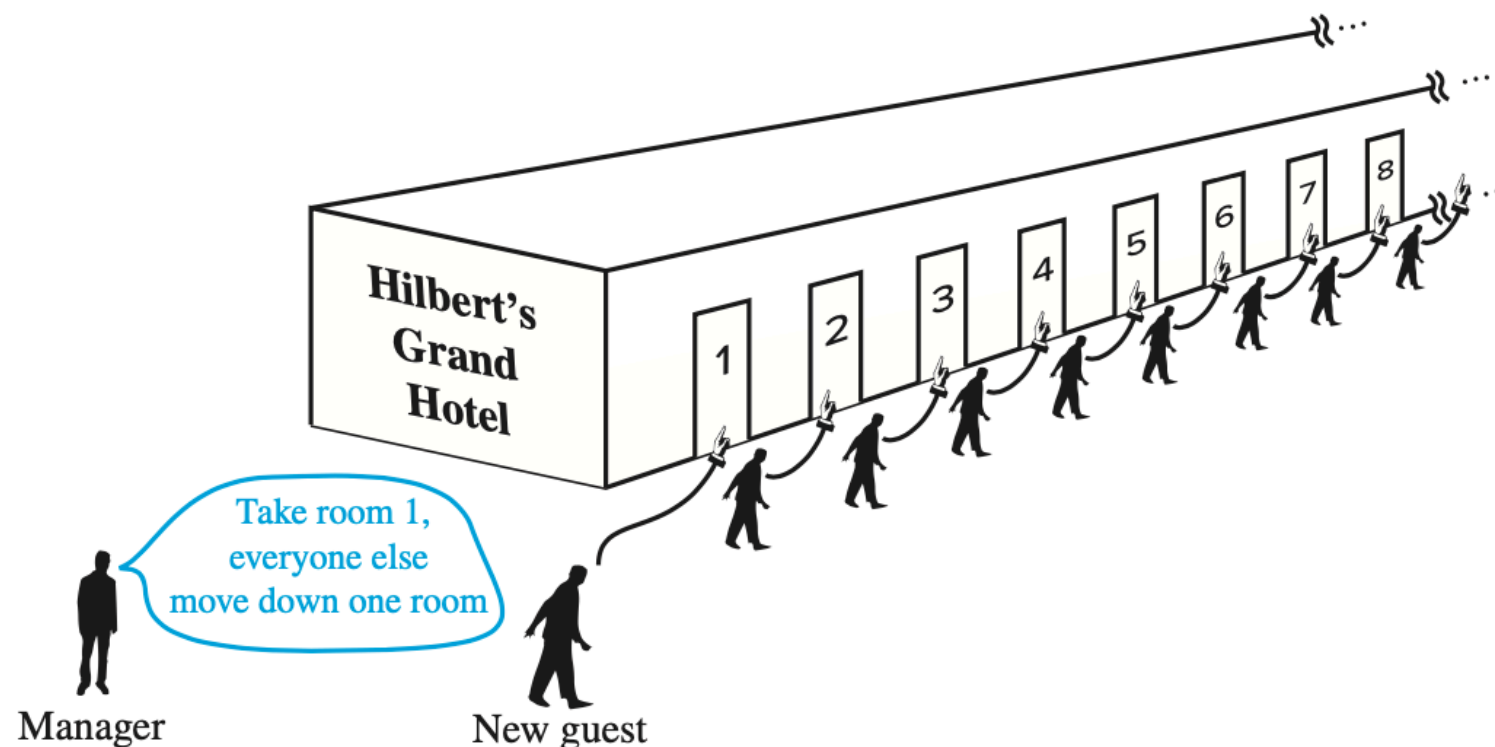
- **Theorem:** If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$. That is, if there are **injective** functions $f : A \rightarrow B$ and $g : B \rightarrow A$, then there exists a **bijective** function between A and B . (Note that **sets A and B can be infinite.**)
 - *The proof is tricky and complicated so omitted here, but you can refer to the textbook [Exercise 41, page 187] if you are interested.*
- Example of its application: show that $|(0, 1)| = |(0, 1]|$
 - Proof:
Construct two one-to-one functions:
 $f : (0, 1) \rightarrow (0, 1], f(x) = x$ * this means $|(0, 1)| \leq |(0, 1]|$
 $g : (0, 1] \rightarrow (0, 1), g(x) = x/2$ * this means $|(0, 1]| \leq |(0, 1)|$

Countable and Uncountable Sets

- **Definition:** A set that is **finite** or has the **same cardinality** as $\mathbb{Z}^+$ is called **countable** 可数, otherwise, it is called **uncountable** 不可数.
 - A countable set S can be **infinite**, but there must exist a **bijection** between $\mathbb{Z}^+$ and S .
- Intuitively, the **cardinality** of a **countable** set is **less than** that of any **uncountable** set. ** formal proof requires the axiom of choice*
- Why the name “**countable**”?
 - All elements in the countable set can be **enumerated and listed** just like listing positive integers 1, 2, 3, ...
 - There exists a list that can **count any element** in a countable set **within finite steps**.

Hilbert's Grand Hotel

- The Grand Hotel has **countably infinite number of rooms**, with **each room occupied by a guest**. We can always **accommodate a new guest** at this hotel.
- This seems impossible because all rooms are already occupied. How can we accommodate the new guest?



- Actually, you can even accommodate **countably many** new guests. How? ** this is left as an assignment problem*

Countable Sets: Example 1

- Example: $A = \{ 0, 2, 4, 6, \dots \}$ * *is this set countable?*
 - (By definition) Is there a **bijection** between $\mathbf{Z}^+$ and A ?
- Proof:
 - Define a function $f : \mathbf{Z}^+ \rightarrow A$, where $x \mapsto 2x - 2$.
 - Show f is a **bijection**!
 - **one-to-one**: if $f(x) = 2x - 2 = 2y - 2 = f(y)$, then $x = y$
 - **onto**: $\forall x \in A$, it has a preimage $(x + 2)/2$ in $\mathbf{Z}^+$
 - Therefore, A is countable.

Countable Sets: Example 2

- **Theorem:** The set of integers $\mathbf{Z}$ is countable.
- Proof:
 - (Directly) List a sequence: $0, 1, -1, 2, -2, 3, -3, \dots$
 - (Alternatively) Define a bijection from $\mathbf{Z}^+$ to $\mathbf{Z}$:
 - when n is even: $f(n) = n/2$
 - when n is odd: $f(n) = -(n - 1)/2$

Countable Sets: Example 3

○ **Theorem:** “The set of rational numbers is countable.”

○ Proof: (rational numbers are of the form p/q)

- List all positive rational numbers:

1. list p/q with $p + q = 2$

$1/1$

2. list p/q with $p + q = 3$

$1/2, 2/1$

3. list p/q with $p + q = 4$

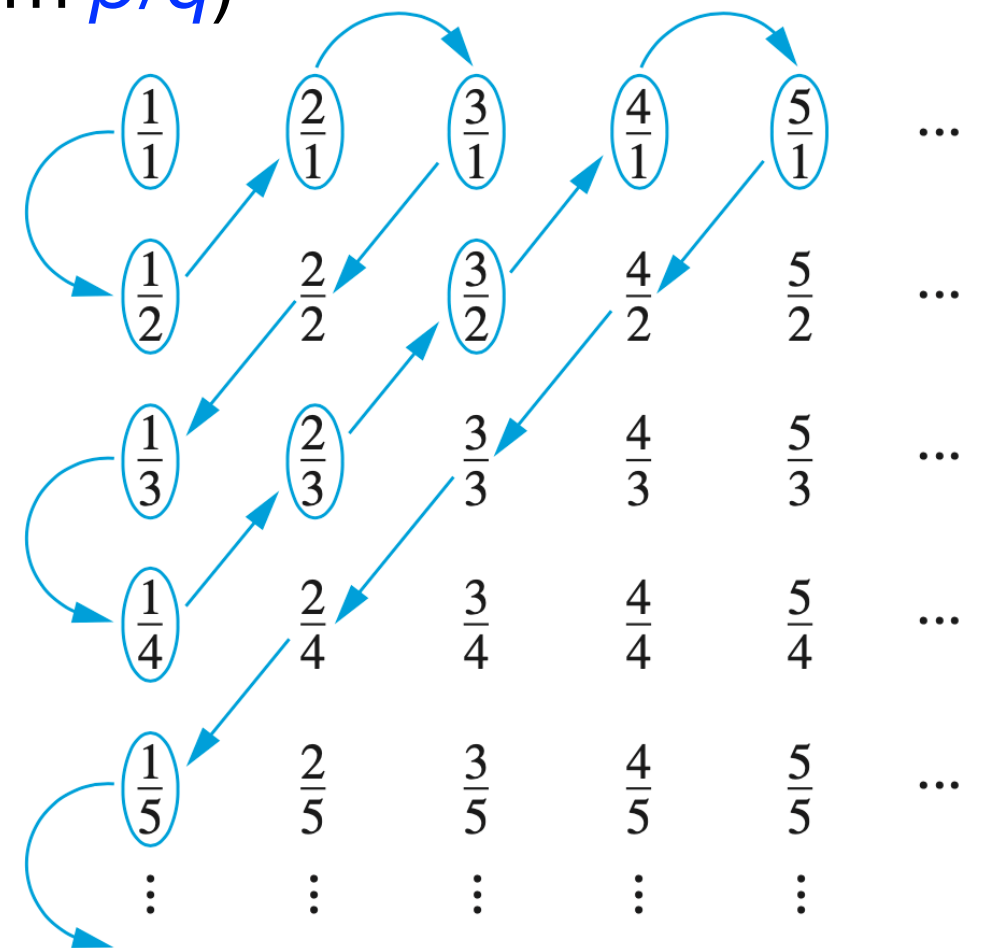
$3/1, \cancel{2/2}, 1/3$

...

- Skip** repeated (uncircled) numbers

- Add 0 and negative numbers to the list

$0, 1, -1, 1/2, -1/2, 2, -2, 3, -3, 1/3, -1/3, \dots$



Countable Sets: Example 4

- **Theorem:** “The set of finite strings S over a finite alphabet A is countable.”
- Proof:
 - Define your favorite alphabetical order for symbols in A
 - We show that the finite strings in S can be listed in a sequence:
 1. list all the strings of length 0 in alphabetical order
 2. list all the strings of length 1 in alphabetical order
 3. list all the strings of length 2 in alphabetical order
 - ...
 - This implies a bijection from $\mathbb{Z}^+$ to S .

Exercise (2 mins)

- Prove that “the set of all Java programs is **countable**.”

- **Theorem:** The set of **finite strings S** over a finite alphabet A is **countable**.
- Proof:
 - Define your favorite alphabetical order for symbols in A
 - We show that the finite strings in S can be listed in a sequence:
 1. list all the strings of length **0** in alphabetical order
 2. list all the strings of length **1** in alphabetical order
 3. list all the strings of length **2** in alphabetical order
 - ...
 - This implies a **bijection from $\mathbb{Z}^+$ to S** .

Exercise (2 mins)

- Prove that “the set of all Java programs is **countable**.”
- Proof:
 - Let **S** be the set of **finite strings** constructed from the **finite alphabet** that consists of all characters that may appear in a Java program. Define any alphabetical order for such characters. Then, as proved in the previous theorem, we can enumerate strings in **S** .
 - For each enumerated string **s** , do the following:
 - feed **s** into a Java compiler
 - if the compiler says YES (i.e., **s** is a syntactically correct Java program), we add **s** to the **list**, otherwise, skip it
 - move on to the next string
 - This implies a **bijection from $\mathbb{Z}^+$ to the set of all Java programs**.

Uncountable Sets: Example

- **Theorem:** “The set of real numbers $\mathbf{R}$ is **uncountable**.”
- Proof by contradiction: (Cantor’s diagonal argument)
 - Assume that $\mathbf{R}$ is **countable**.
Then, **every subset of $\mathbf{R}$ is countable**. * *left as an assignment problem*
In particular, interval $[0, 1]$ is **countable**. This implies that there exists a list $r_1, r_2, r_3, \dots$ that can enumerate all elements in $[0, 1]$, where
$$r_1 = 0.\mathbf{d_{11}}d_{12}d_{13}d_{14} \dots$$
$$r_2 = 0.d_{21}\mathbf{d_{22}}d_{23}d_{24} \dots$$
$$r_3 = 0.d_{31}d_{32}\mathbf{d_{33}}d_{34} \dots$$
$$\dots$$
with $d_{ij} \in \{0, 1, 2, \dots, 9\}$ * *note that $1 = 0.999999\dots$*
 - Construct a real number r that is **not included** in the above list:
$$r = 0.d_1d_2d_3d_4 \dots \quad \text{where } d_i \neq \mathbf{d_{ii}}$$

Exercise (2 mins)

- Prove that “the power set $\mathcal{P}(\mathbf{N})$ is **uncountable**.”

Recall that $\mathcal{P}(\mathbf{N})$ contains all subsets of $\mathbf{N}$

- **Theorem:** “The set of real numbers $\mathbf{R}$ is **uncountable**.”
- Proof by contradiction: (Cantor’s diagonal argument)
 - Assume that $\mathbf{R}$ is **countable**.
Then **every subset of $\mathbf{R}$ is countable**. In particular, the interval $[0, 1]$ is **countable**. This implies that there exists a list $r_1, r_2, r_3, \dots$ that can enumerate all elements in $[0, 1]$, where
$$r_1 = 0.\textcolor{blue}{d}_{11}\textcolor{red}{d}_{12}d_{13}d_{14} \dots$$
$$r_2 = 0.d_{21}\textcolor{red}{d}_{22}d_{23}d_{24} \dots$$
$$r_3 = 0.d_{31}d_{32}\textcolor{red}{d}_{33}d_{34} \dots$$
$$\dots$$
with $d_{ij} \in \{0, 1, 2, \dots, 9\}$ *Note that $1 = 0.999999\dots$*
 - Construct a real number r that is **not included** in the above list:
$$r = 0.d_1d_2d_3d_4 \dots \quad \text{where } d_i \neq \textcolor{red}{d}_{ii}$$

Exercise (2 mins)

- Prove that “the power set $\mathcal{P}(\mathbf{N})$ is **uncountable**.”
- Proof by contradiction: (Cantor’s diagonal argument)
 - Assume that $\mathcal{P}(\mathbf{N})$ is **countable**.
This means that all elements of $\mathcal{P}(\mathbf{N})$ can be listed as $S_0, S_1, S_2, \dots$ where $S_i \in \mathcal{P}(\mathbf{N})$. Then, each $S_i \subseteq \mathbf{N}$ can be represented by a bit string $b_{i0}b_{i1}b_{i2}\dots$, where $b_{ij} = 1$ if $j \in S_i$ and $b_{ij} = 0$ if $j \notin S_i$:

$$S_0 = \mathbf{b_{00}}b_{01}b_{02}b_{03} \dots$$

$$S_1 = b_{10}\mathbf{b_{11}}b_{12}b_{13} \dots$$

$$S_2 = b_{20}b_{21}\mathbf{b_{22}}b_{23} \dots$$

$$\dots \text{ with } b_{ij} \in \{0, 1\}$$

- Construct a set $S \in \mathcal{P}(\mathbf{N})$ that is **not included** in the above list:

$$S = b_0b_1b_2b_3 \dots \quad \text{where } b_i \neq \mathbf{b_{ii}}$$

Computable vs Uncomputable

- **Definition:** We say that a function is **computable** 可计算 if there is a computer program in some programming language that finds the values of this function. If a function is **not** computable, we say it is **uncomputable** 不可计算.
- **Theorem:** There exist uncomputable functions. ** very cool!*
- Proof sketch:
 - **Part 1:** The set of all computer programs in all programming languages is **countable**. ** left as an assignment problem*
 - **Part 2:** The set of all functions from $\mathbf{Z}^+$ to $\{0, 1, \dots, 9\}$ is **uncountable**. ** left as an assignment problem*
 - **Conclusion:** there exists a function $f^* : \mathbf{Z}^+ \rightarrow \{0, 1, \dots, 9\}$ that cannot be computed by any computer program, i.e., f^* is **uncomputable**. ** proof by contradiction*

The Continuum Hypothesis

- We know that $|\mathbf{N}| < |\mathcal{P}(\mathbf{N})|$, because $|\mathbf{N}| \leq |\mathcal{P}(\mathbf{N})|$ and $\mathbf{N}$ is countable while $\mathcal{P}(\mathbf{N})$ is uncountable.
 - **Cantor's Theorem:** $|S| < |\mathcal{P}(S)|$ holds for any set S
- Question: Is there a set A such that $|\mathbf{N}| < |A| < |\mathcal{P}(\mathbf{N})|$?
- **Continuum Hypothesis:** The above set A **does not exist**.
 - This is a very important open problem in mathematics.

04 Complexity of Algorithms

To be continued...

Quiz 1

- Quiz 1 will take place in class on Oct 14 and it captures materials from 02 Logic and Proofs to 03 Sets and Functions.
- We will have 2 open-book quizzes in total for this course:
 - 3~6 questions in ~20 minutes for each quiz.
 - Must attend the quizzes in person and write in English.
 - Please bring several pieces of paper to write your answers on.
 - No electronic device is allowed during the quiz.
 - After quiz, please take photos of your quiz answers and submit to Blackboard (you will have ~5 minutes after quiz to do this).